4DVarNet-FM Synthesis Report

4DVarNet-FM: Experiment Synthesis Report (live-updated)

Generated: 2026-06-29 18:11:09 | T_max = 3.0s | dt = 0.01 | 15 obs/window

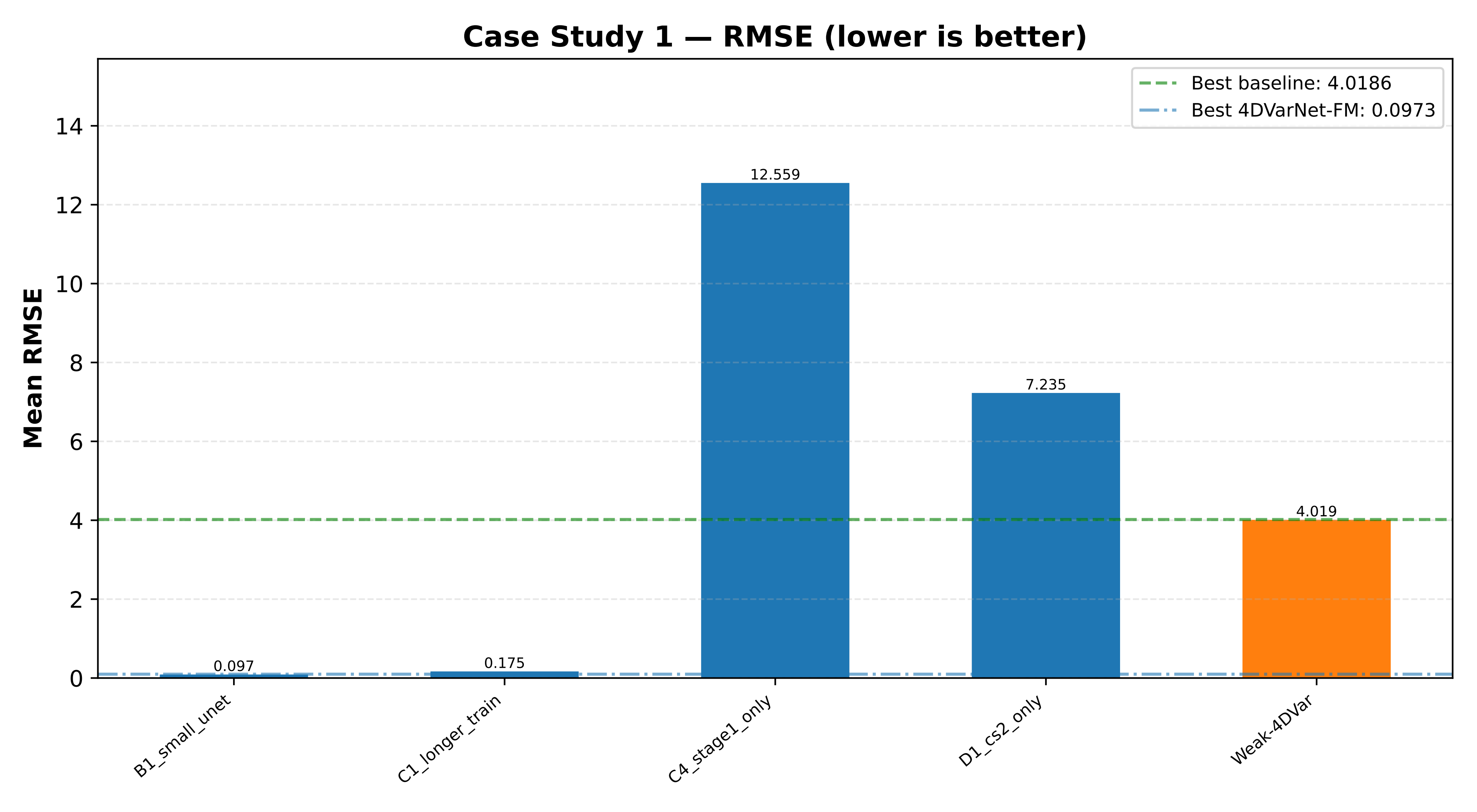
Part 1 — 4DVarNet-FM Variants:

Config	CS1 X	CS1 Y	CS1 Z	CS1 μ	CS2 X	CS2 Y	CS2 Z	CS2 μ	Deg	Time	
B1_small_unet	0.081	0.098	0.113	0.097	0.100	0.109	0.122	0.110	1.13	x 47m	◄
C1_longer_train	0.172	0.165	0.190	0.175	0.178	0.166	0.192	0.179	1.02	x 105m	
C4_stage1_only	5.239	9.310	23.128	12.559	5.340	9.312	22.558	12.404	0.99	x 5m	
D1_cs2_only	7.754	6.510	7.441	7.235	7.247	6.606	7.598	7.150	0.99	x 33m	

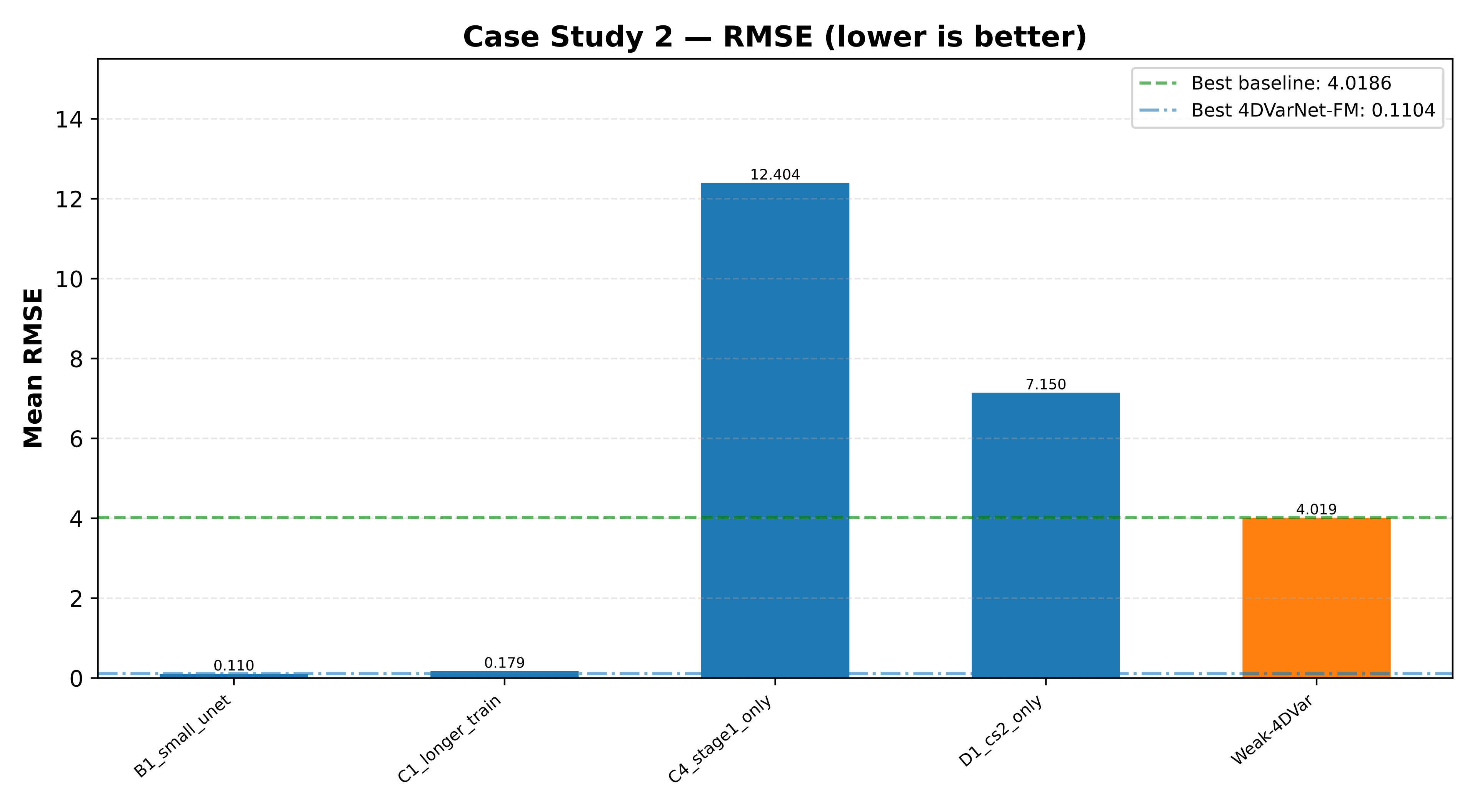
Part 2 — DA Baselines (same test data):

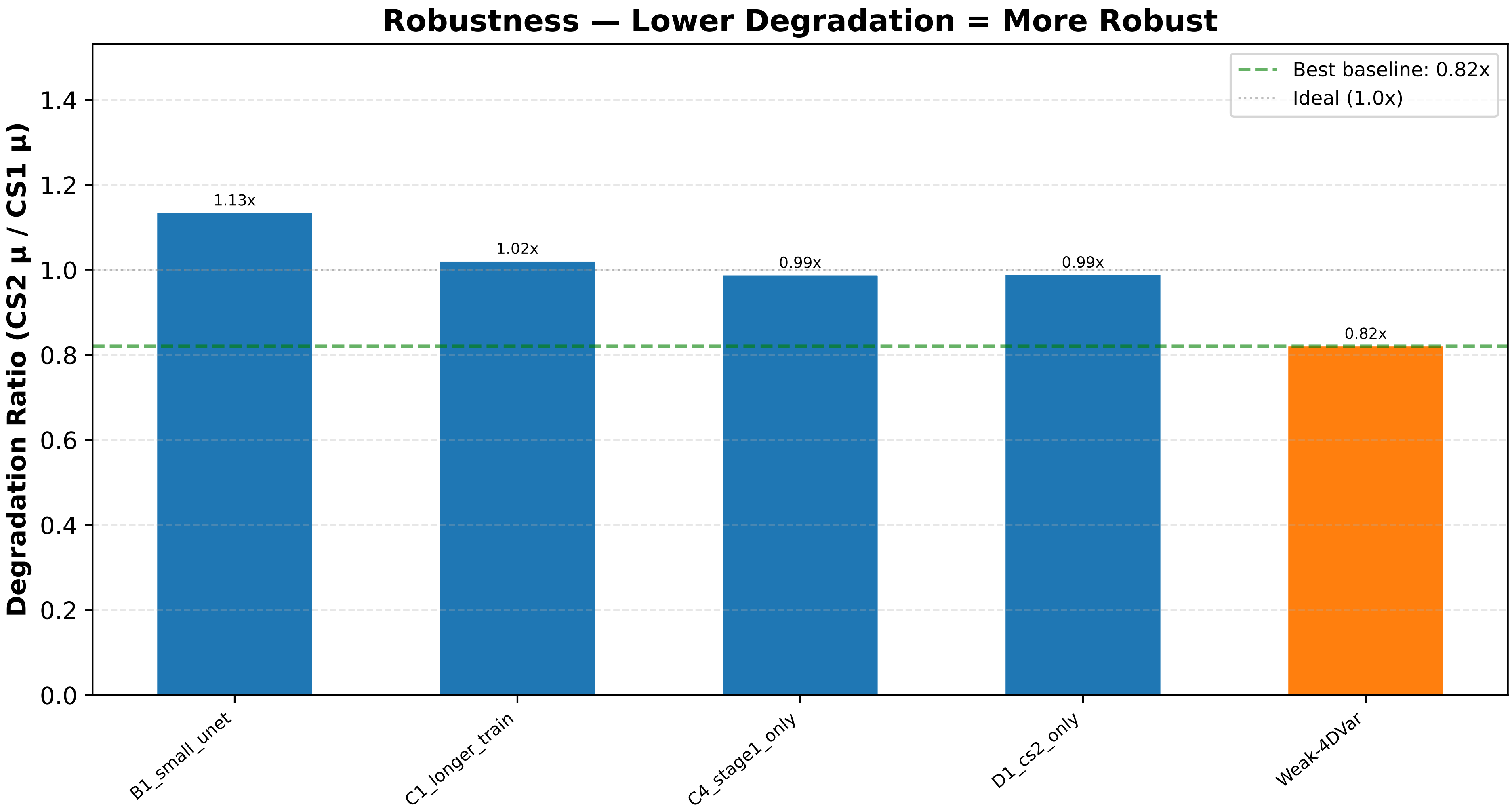
Method	CS1 μ	CS2 μ	Deg	
Weak-4DVar	4.8970	4.0186	0.82	x

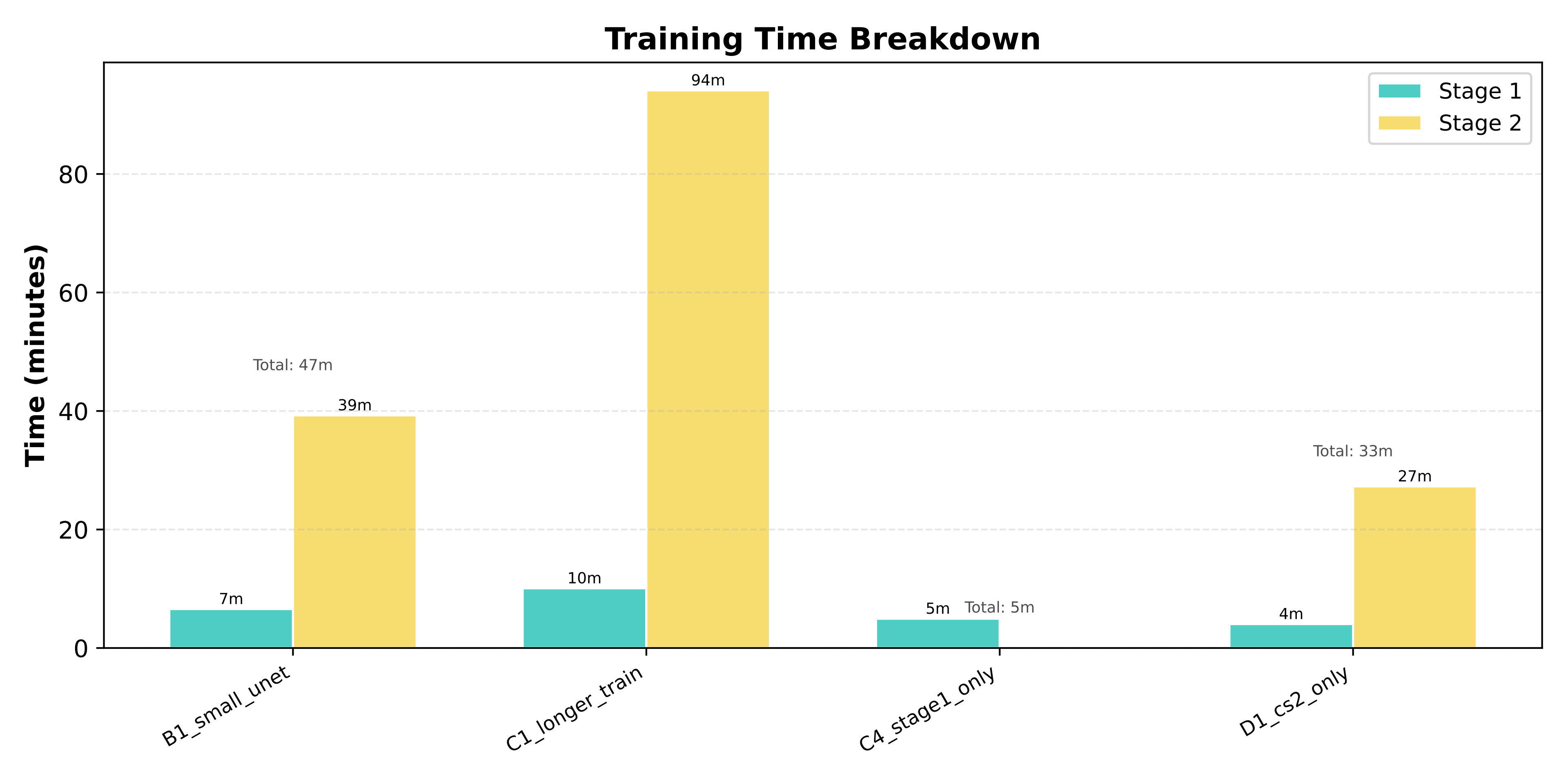
Best 4DVarNet-FM: B1_small_unet (CS2 μ=0.1104)
Best baseline CS1: EnKF μ=1.2699
Best baseline CS2: Weak-4DVar μ=4.0186



Case Study 1 — RMSE (lower is better)
Best baseline: 4.0186
Best 4DVarNet-FM: 0.0973
Mean RMSE
0
2
4
6
8
10
12
14
0.097
0.175
12.559
7.235
4.019
B1_small_unet
C1_longer_train
C4_stage1_only
D1_cs2_only
Weak-4DVar


Case Study 2 — RMSE (lower is better)
Best baseline: 4.0186
Best 4DVarNet-FM: 0.1104
Mean RMSE
0
2
4
6
8
10
12
14
0.110
0.179
12.404
7.150
4.019
B1_small_unet
C1_longer_train
C4_stage1_only
D1_cs2_only
Weak-4DVar


Robustness — Lower Degradation = More Robust
Best baseline: 0.82x
Ideal (1.0x)
Degradation Ratio (CS2 μ / CS1 μ)
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
1.13x
1.02x
0.99x
0.99x
0.82x
B1_small_unet
C1_longer_train
C4_stage1_only
D1_cs2_only
Weak-4DVar



Training Time Breakdown
Stage 1
Stage 2
Time (minutes)
0
20
40
60
80
Total: 47m
7m
39m
10m
94m
5m
Total: 5m
4m
27m
Total: 33m
B1_small_unet
C1_longer_train
C4_stage1_only
D1_cs2_only

4DVarNet-FM Synthesis Report — Conclusion

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Conclusion & Recommendations  (live-updated)
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Best 4DVarNet-FM variant:  B1_small_unet
  CS1 mean RMSE:  0.0973  (best baseline EnKF: 1.2699)  ✓ BEATS baseline
  CS2 mean RMSE:  0.1104  (best baseline Weak-4DVar: 4.0186)  ✓ BEATS baseline
  Degradation:    1.13x

Comparison across all variants:
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  B1_small_unet         CS1 μ=0.0973    CS2 μ=0.1104    Deg=1.13x  ★ BEATS ALL BASELINES
  C1_longer_train       CS1 μ=0.1753    CS2 μ=0.1788    Deg=1.02x  ★ BEATS ALL BASELINES
  C4_stage1_only        CS1 μ=12.5590   CS2 μ=12.4035   Deg=0.99x  Below baselines
  D1_cs2_only           CS1 μ=7.2349    CS2 μ=7.1504    Deg=0.99x  Below baselines

Key Insights:
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Recommendation:
  Use B1_small_unet for production (lowest CS2 RMSE).
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